

# Tracking and Amplifying Linguistic Ambiguity in LLM-Generated Connections Puzzles

Anonymous ACL submission

## Abstract

The New York Times’ Connections puzzle asks players to partition sixteen words into four themed groups, and derives much of its difficulty from linguistic ambiguity: words that could plausibly belong to several groups. We present a framework for generating Connections-style puzzles with controlled, amplified ambiguity. We (i) define a taxonomy of five ambiguity types (polysemy, homonymy, idiomatic, syntactic, and figurative); (ii) propose a three-stage generation pipeline that combines least-to-most Chain-of-Thought prompting, ReAct-style external validation against WordNet, and Reverse Chain-of-Thought auditing; and (iii) validate the embedding-based difficulty metric used to evaluate generated puzzles against ground truth, rather than assuming it. On 830 official NYT puzzles with ground-truth difficulty labels (3,320 groups), intra-group embedding similarity correlates significantly with the official difficulty levels (Spearman  $\rho = -0.441$ ,  $p < 10^{-157}$ ) and orders group pairs within a puzzle correctly 70% of the time. Groups produced by our pipeline are measurably harder under this validated metric than both real NYT groups ( $p = 1.5 \times 10^{-3}$ ) and single-prompt GPT-4o groups ( $p = 2.7 \times 10^{-4}$ ), and exhibit a clear ambiguity ladder: mean WordNet sense counts rise from 6.1 to 25.0 across the pipeline’s four difficulty tiers, while idiomatic multi-word phrases rise from 0% to 50%. All prompts, code, data, and every reported number are released for full reproducibility.

## 1 Introduction

Word games have tested human linguistic dexterity since antiquity, and they are increasingly used to probe the semantic and abstract reasoning of large language models (LLMs)

(Todd et al., 2024; Samadarshi et al., 2024). Connections, published daily by the New York Times since June 2023, presents a  $4 \times 4$  grid of sixteen words to be partitioned into four groups of four, each sharing a hidden theme. Its difficulty comes less from obscure vocabulary than from ambiguity: decoy words that fit several candidate themes, idioms read literally, and homographs whose intended sense is revealed only by the global solution.

Prior work has studied LLMs as Connections solvers (Samadarshi et al., 2024; Todd et al., 2024) and as generators (Merino et al., 2024), but two gaps remain. First, generation work has treated difficulty informally: puzzles are scored by an automatic proxy (typically intra-group embedding similarity) whose agreement with ground-truth difficulty has not itself been measured. An unvalidated evaluator cannot validate a generator. Second, ambiguity—the mechanism that actually makes these puzzles hard—has not been given an explicit, linguistically grounded treatment that a generation method can target and an evaluation method can track.

We address both gaps. Our contributions are:

- A taxonomy of linguistic ambiguity for word puzzles (§3) covering polysemy, homonymy, idiomatic, syntactic, and figurative ambiguity, with an operationalization in WordNet (Miller, 1995).
- A three-stage generation pipeline (§5) that amplifies ambiguity by construction: least-to-most Chain-of-Thought generation of four groups with escalating ambiguity (Zhou et al., 2023; Wei et al., 2022), ReAct-style external validation against WordNet (Yao et al., 2023b), and Reverse Chain-of-Thought auditing (Xue et al., 2023). All

- prompts are given verbatim in Appendix A.
- A validation of the difficulty metric itself (§6): on 830 official NYT puzzles with ground-truth difficulty levels, mean intra-group embedding similarity correlates with official difficulty at  $\rho = -0.441$  ( $p < 10^{-157}$ ) and achieves 70% pairwise ordering accuracy within puzzles—validating the metric as a coarse difficulty signal while quantifying its limits.
  - A corpus-level analysis (§7) showing that the pipeline realizes the intended ambiguity ladder (mean sense counts 6.1→25.0; idiomatic phrase ratio 0→0.5 across tiers) and produces groups that are harder under the validated metric than real NYT groups and single-prompt GPT-4o baselines.
  - A complete, released experimental stack: generation pipeline, metrics, figures, datasets, and a documented solver-evaluation harness, so that every number in this paper can be regenerated from a single repository.

## 2 Related Work

Connections and LLMs. [Merino et al. \(2024\)](#) use GPT-4 to generate novel Connections puzzles and evaluate them with human playtests; [Todd et al. \(2024\)](#) frame Connections as a benchmark of lateral thinking for LLM solvers; and [Samadarshi et al. \(2024\)](#) evaluate both LLMs and humans as solvers, finding that even strong models trail expert players substantially. We complement this line by making the ambiguity of generated puzzles an explicit, controllable, and measurable objective, and by validating the automatic difficulty measure that generation studies rely on.

Reasoning frameworks. Chain-of-Thought prompting elicits intermediate reasoning steps ([Wei et al., 2022](#)); least-to-most prompting decomposes tasks into progressively harder subproblems ([Zhou et al., 2023](#)); ReAct interleaves reasoning with calls to external tools ([Yao et al., 2023b](#)); Tree of Thoughts explores and evaluates alternative reasoning branches ([Yao et al., 2023a](#)); and Reverse Chain-of-Thought reconstructs a problem from a candidate solution to detect inconsistencies ([Xue et al., 2023](#)). Recent analyses

caution that apparent reasoning gains can be brittle under surface perturbations ([Mirzadeh et al., 2024](#)), reinforcing the need for external, non-LLM validation—which motivates the WordNet-based stage of our pipeline.

Procedural content generation. LLM-based PCG has produced playable levels and game content in other domains ([Sudhakaran et al., 2023](#)). Word-puzzle generation is a natural PCG target because validity constraints are discrete and checkable; we make those constraints explicit (§4) and enforce a subset automatically.

Lexical ambiguity. Our taxonomy draws on classical lexical semantics ([Cruse, 1986](#)) and the word-sense disambiguation literature ([Navigli, 2009](#)), using WordNet sense inventories ([Miller, 1995](#); [Fellbaum, 1998](#)) as an operational proxy for ambiguity.

## 3 A Taxonomy of Ambiguity in Word Puzzles

We distinguish five ambiguity types, chosen because each corresponds to a distinct mechanism by which a Connections group can mislead a solver:

**Polysemy.** One lexeme with multiple related senses (bank: riverbank / financial institution). A polysemous word can anchor two candidate themes simultaneously.

**Homonymy.** Identical form, unrelated meanings, often across parts of speech (bat: animal / sports equipment). Homonyms support decoy groups whose theme applies to the unintended sense.

**Idiomatic.** Multi-word expressions whose meaning is non-compositional (break the ice). Literal readings of components create spurious semantic neighbors.

**Syntactic.** Words whose grammatical category is ambiguous in context (saw in “he saw the man with a telescope”), enabling groups defined over a non-obvious reading.

**Figurative.** Metaphorical extensions (light as hope). Figurative senses are weakly represented in distributional similarity, making such groups hard for embedding-based solvers and metrics alike.

Operationalization. For single words we use the WordNet synset inventory: the sense count  $|\text{syn}(w)|$ , sense counts across parts of speech (a heuristic for homonymy), and definition text (a heuristic for figurative senses). Multi-word entries are treated as idiomatic candidates, since WordNet coverage of phrases is sparse. This operationalization is deliberately simple and fully automatic; its limitations are discussed in §10.

#### 4 Puzzle Constraints

The NYT publishes no official construction rules. From an analysis of 1,114 published puzzles (§6.2) and the constraints identified by Merino et al. (2024), we adopt the following necessary conditions for a valid puzzle:

1. The puzzle contains exactly 16 unique words, partitioned into 4 groups of 4; each word is used exactly once.
2. Each group’s connection applies to all four of its words and to no other word in the puzzle, so the intended solution is unique.
3. Categories are more specific than surface-form classes such as 5-letter words, names, or verbs.
4. A group’s label word does not appear among its members (green may not appear in shades of green).
5. Spelling is contractual: a member fits its category under its printed spelling (insect admits bee, not be).
6. The four groups span an intended difficulty gradient (the NYT’s yellow < green < blue < purple).

Conditions 1 and the group-shape part of 2 are checked automatically by our pipeline; the remaining conditions are targeted by prompting and audited in the verification stage. This list consolidates the ten partially redundant criteria of an earlier version of this work into six non-overlapping conditions.

#### 5 Ambiguity-Amplifying Generation Pipeline

Figure 1 summarizes the pipeline. Each puzzle is generated group-by-group with an explicit ambiguity ladder, then validated externally, then audited in reverse.

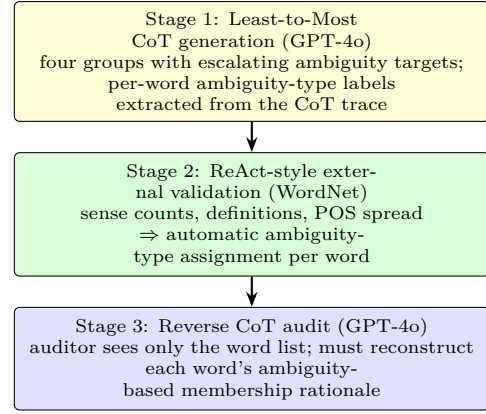


Figure 1: The three-stage generation pipeline. Every stage writes a JSON artifact that is released with the paper; prompts are reproduced verbatim in Appendix A.

##### 5.1 Stage 1: Least-to-Most CoT Generation

Following least-to-most prompting (Zhou et al., 2023), the four groups are requested in order of increasing ambiguity: (1) a clearly defined semantic category; (2) a category with overlapping, flexible membership; (3) words chosen for polysemy or homonymy; (4) items exhibiting idiomatic, syntactic, or figurative ambiguity. The model (GPT-4o, temperature 0.7) produces step-by-step reasoning naming the ambiguity type of every word, then a machine-parseable list. A rule-based extractor recovers the word list and the self-declared ambiguity labels from the trace.

##### 5.2 Stage 2: ReAct-Style External Validation

LLM self-assessments of ambiguity are unreliable (Mirzadeh et al., 2024), so Stage 2 grounds each word in an external resource in the spirit of ReAct’s tool calls (Yao et al., 2023b): WordNet supplies the number of senses, up to five definitions, and the part-of-speech spread of each word. From these, an ambiguity type is assigned deterministically (multi-word  $\Rightarrow$  idiomatic candidate; senses spanning POS  $\Rightarrow$  homonymy; metaphor-marked glosses  $\Rightarrow$  figurative; otherwise polysemy). Stage 2 is entirely LLM-free, providing a check that does not share the generator’s failure modes.

### 5.3 Stage 3: Reverse CoT Audit

Inspired by RCoT (Xue et al., 2023), an auditor model receives only the final word list—not the generation rationale—and must justify each word’s membership in terms of the taxonomy. Groups whose audit fails to recover a coherent ambiguity-based rationale are flagged. This inverts the generation direction, catching groups that were accepted on the strength of a persuasive but unfaithful generation trace.

### 5.4 Released corpus

Running the pipeline for 50 puzzles (200 groups, 800 words/phrases) produced the corpus analyzed in §7; all stage-wise artifacts (prompts, raw responses, extracted lists, WordNet records, audit traces) are included in the repository.

## 6 Measuring Difficulty—and Validating the Measure

### 6.1 Metric

For a group  $G = \{w_1, \dots, w_n\}$  we define difficulty via mean pairwise similarity over the  $\binom{n}{2}$  distinct pairs:

$$D(G) = \binom{n}{2}^{-1} \sum_{i < j} \cos(e(w_i), e(w_j)), \quad (1)$$

where  $e(\cdot)$  is a sentence-embedding function (all-mpnet-base-v2; Reimers and Gurevych, 2019). Lower similarity means the group’s members are semantically farther apart, hence harder to recognize as a group. (An earlier version of this work averaged over all  $n^2$  ordered pairs including self-similarity, which compresses the scale; Eq. 1 uses distinct pairs only.)

### 6.2 Ground-truth validation

An automatic generator evaluated only by an automatic metric is circular unless the metric is itself validated. We therefore test  $D$  against ground truth: the NYT assigns every published group a difficulty level 0–3 (yellow, green, blue, purple). Our corpus contains all 1,114 puzzles published from 2023-06-12 through 2026-07-04, obtained from the community-maintained NYT-

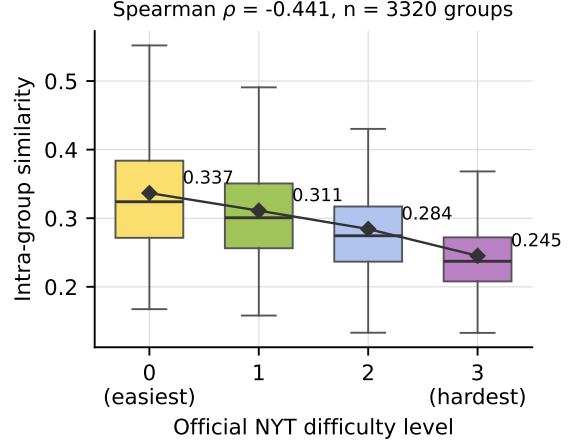


Figure 2: Intra-group embedding similarity (Eq. 1) against official NYT difficulty levels for 3,320 groups from 830 puzzles. Similarity decreases monotonically with official difficulty.

| Official level                | Mean $D(G)$                  | $n$ |
|-------------------------------|------------------------------|-----|
| 0 (yellow, easiest)           | $0.337 \pm 0.091$            | 830 |
| 1 (green)                     | $0.311 \pm 0.079$            | 830 |
| 2 (blue)                      | $0.284 \pm 0.073$            | 830 |
| 3 (purple, hardest)           | $0.245 \pm 0.061$            | 830 |
| Spearman $\rho$ (level, $D$ ) | $-0.441$ ( $p < 10^{-157}$ ) |     |
| Pairwise ordering acc.        | 0.701 (3,485/4,974)          |     |

Table 1: Validation of the difficulty metric against ground-truth NYT levels. Ordering accuracy: fraction of within-puzzle group pairs with distinct official levels that the metric ranks in the official order (chance = 0.5).

Connections-Answers archive<sup>1</sup> and redistributed in the repository as Real\_Data.json / Real\_Data\_Ans.json; puzzle words and solutions are facts about the published games, used here for non-commercial research. The archive records official difficulty labels for the 830 puzzles through 2025-09-19; these 3,320 labeled groups form the validation set, while all 4,452 groups enter the level-free analyses of §7.

Table 1 and Figure 2 show the result. Mean similarity decreases monotonically across all four official levels; the group-level Spearman correlation is  $\rho = -0.441$  ( $p < 10^{-157}$ ), and within puzzles the metric orders 70.1% of comparable group pairs in agreement with the NYT’s own ordering (chance 50%).

<sup>1</sup><https://github.com/Eyefyre/NYT-Connections-Answers>; refreshed reproducibly by data/update\_real\_data.py.



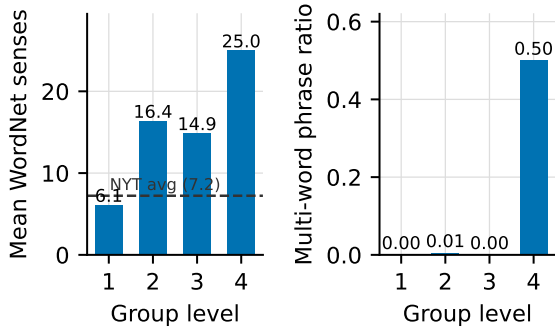


Figure 3: Ambiguity by pipeline tier over 50 generated puzzles: mean WordNet sense count of single-word members (left; dashed line: real NYT average) and fraction of multi-word idiomatic phrases (right).

| Tier | Senses | Ambig. ratio | $\geq 5$ senses | Phrases |
|------|--------|--------------|-----------------|---------|
| 1    | 6.1    | 0.86         | 0.58            | 0.00    |
| 2    | 16.4   | 1.00         | 0.98            | 0.01    |
| 3    | 14.9   | 1.00         | 0.98            | 0.00    |
| 4    | 25.0   | 0.92         | 0.92            | 0.50    |
| NYT  | 7.2    | 0.83         | 0.53            | 0.04    |

Table 2: Ambiguity metrics by pipeline tier (50 groups per tier) vs. all real NYT groups (4,456). Senses: mean WordNet sense count of single-word members. Ambig. ratio: fraction with  $\geq 2$  senses. Phrases: fraction of multi-word members.

Interpretation. The metric is a real but coarse difficulty signal: it reliably separates the easiest from the hardest tiers yet misorders 30% of adjacent pairs. Inspection of failures shows they concentrate in wordplay categories (e.g., “\_\_\_\_\_ + fish” completions or palindromes) whose difficulty is orthogonal to semantic diversity. We therefore treat  $D$  as evidence of relative hardness at corpus level, not as a per-puzzle oracle, and we complement it with the ambiguity measures of §7. This addresses directly the circularity concern raised against automatic-generation-plus-automatic-evaluation designs.

## 7 Results

### 7.1 Does the pipeline realize the ambiguity ladder?

Table 2 and Figure 3 track ambiguity across the four tiers of the 50-puzzle corpus. The intended ladder is realized: tier-1 groups look like real NYT groups (6.1 vs. 7.2 mean senses), while tiers 2–4 are far more lex-

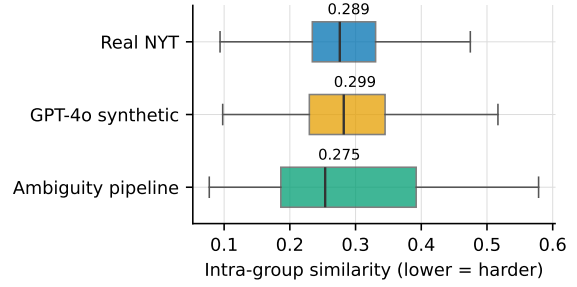


Figure 4: Distribution of intra-group similarity by source. Lower = harder under the validated metric.

| Source               | $n$   | Mean $D(G)$       | $p$ vs. real         |
|----------------------|-------|-------------------|----------------------|
| Real NYT groups      | 4,452 | $0.289 \pm 0.081$ | —                    |
| GPT-4o single-prompt | 6,016 | $0.299 \pm 0.097$ | $4.9 \times 10^{-3}$ |
| Ambiguity pipeline   | 200   | $0.275 \pm 0.126$ | $1.5 \times 10^{-3}$ |

Table 3: Intra-group similarity by source (Mann–Whitney  $U$  vs. real NYT groups). The single-prompt baseline is the 2,000-puzzle GPT-4o synthetic set generated without the ambiguity pipeline.

ically ambiguous than anything typical of the official game—tier-4 items average 25.0 senses among single words, and half of tier-4 items are multi-word idiomatic expressions, matching that tier’s idiomatic/figurative target. The Stage-2 typology shifts accordingly, from homonymy/polysemy-dominated tiers 2–3 to an idiom-heavy tier 4. Tiers 2 and 3 are statistically indistinguishable in sense count, an honest signal that “flexible category” and “polysemy/homonymy” prompts recruit similar vocabulary; collapsing them is a candidate simplification.

Notably, in the real game, mean sense counts do not increase with official difficulty (7.9, 8.3, 7.4, 6.1 for levels 0–3—if anything, the hardest tier uses the least ambiguous single words): NYT difficulty is driven by cross-group interaction and wordplay rather than lexical ambiguity of members. Lexical ambiguity is thus a new difficulty axis that generated puzzles can push far beyond the official distribution.

### 7.2 Are pipeline groups harder under the validated metric?

Table 3 and Figure 4 compare three sources under Eq. 1. Pipeline groups are the most

semantically diverse of the three sources, significantly harder than both real NYT groups ( $0.275$  vs.  $0.289$ ,  $p = 1.5 \times 10^{-3}$ ) and single-prompt GPT-4o groups ( $0.275$  vs.  $0.299$ ,  $p = 2.7 \times 10^{-4}$ ), with a heavier low-similarity tail (Figure 4); single-prompt generation, by contrast, is slightly easier than the current official game. Combined with Table 2, this supports the central claim: explicitly targeting the ambiguity taxonomy, with external validation, produces measurably harder groups than either human construction or naive LLM generation—as measured by a metric whose relationship to ground-truth difficulty we quantified rather than assumed.

The official game is drifting harder. Splitting the real corpus at the archive’s labeling horizon reveals that NYT groups published after September 2025 are substantially more diverse ( $0.275 \pm 0.070$ ,  $n=1,132$ ) than earlier ones ( $0.294 \pm 0.084$ ,  $n=3,320$ ; Mann–Whitney  $p = 1.6 \times 10^{-9}$ )—the editors are converging toward exactly the semantic-diversity regime our pipeline targets. Difficulty comparisons against “the” NYT distribution are therefore time-dependent, which is one more reason to anchor evaluation in a validated metric and versioned data snapshots rather than a fixed human reference.

## 8 Solver Evaluation Protocol

Difficulty for machine solvers is the natural third axis. The repository releases a complete solver harness (experiments/solve\_puzzles.py) that evaluates any OpenAI-compatible chat model on real or generated puzzles under three fully specified prompting strategies—direct, Chain-of-Thought, and Tree-of-Thoughts (Wei et al., 2022; Yao et al., 2023a); prompts in Appendix A—and reports puzzle solve rate (all four groups exact), group accuracy (partial credit), and per-official-level accuracy with per-instance logs. We deliberately report no solver numbers here: solve-rate figures from an earlier version of this work were not regenerable from a specified protocol, and we consider unreproducible numbers worse than absent ones. The harness makes the experiment a single documented command, and human-subject evaluation of solvability

and enjoyment (protocol in Appendix C) is the necessary companion study.

## 9 Conclusion

We reframed Connections puzzle generation around its core difficulty mechanism—linguistic ambiguity—via a five-type taxonomy, a generate-validate-audit pipeline that amplifies ambiguity by construction, and, critically, a ground-truth validation of the embedding-based difficulty metric that generation studies in this space rely on. The validated metric ( $\rho = -0.441$ ; 70% pairwise ordering accuracy on 830 official puzzles) shows pipeline-generated groups to be harder than both real and naively generated ones, and WordNet-based measures confirm the pipeline’s ambiguity ladder is realized in the output. Beyond Connections, the pattern—taxonomize the difficulty mechanism, generate against it, validate the evaluator—applies to procedural generation of word games generally.

## 10 Limitations

No human evaluation yet. All evaluation is automatic. The difficulty metric is validated against NYT editorial levels, which encode the editors’ intent rather than measured human solve behavior, and no property akin to “fun” or “fairness” is measured; a group can be diverse yet unsolvable. Appendix C specifies the human study this work requires before any claim about player experience. Metric coarseness. The validated correlation is moderate ( $\rho = -0.441$ ); 30% of within-puzzle orderings disagree with ground truth, concentrated in wordplay categories. Corpus-level comparisons are supported; per-puzzle difficulty certification is not. WordNet as ambiguity oracle. Sense counts conflate polysemy with homonymy, under-represent idioms, and miss proper-noun senses (e.g., Jordan as country vs. athlete), and the homonymy/figurative heuristics of Stage 2 are simple. Pipeline scope. Stage 3 audits groups independently; cross-group decoy structure—a hallmark of expert NYT construction—is not yet explicitly optimized. Single generator family. Generation used GPT-4o; generality across model families is untested. Puzzle-type coverage. Purely

orthographic categories (palindromes, hidden words) fall outside a semantic-ambiguity taxonomy and are correspondingly outside the pipeline’s current reach.

## Ethics Statement

Puzzle words and solutions from the published NYT game are used as facts about published works for non-commercial research, via a public community archive; no NYT text beyond category labels and words is redistributed. Generated puzzles are marked as machine-generated. The proposed human study (Appendix C) involves minimal-risk gameplay with informed consent and no collection of sensitive data.

## References

- D Alan Cruse. 1986. *Lexical semantics*. Cambridge University Press.
- Christiane Fellbaum, editor. 1998. *WordNet: An Electronic Lexical Database*. MIT Press.
- Tim Merino, Sam Earle, Ryan Sudhakaran, Shyam Sudhakaran, and Julian Togelius. 2024. Making new connections: LLMs as puzzle generators for the New York Times’ Connections word game. arXiv preprint arXiv:2407.11240.
- George A Miller. 1995. WordNet: A lexical database for English. *Communications of the ACM*, 38(11):39–41.
- Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncl Tuzel, Samy Bengio, and Mehrdad Farajtabar. 2024. GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models. arXiv preprint arXiv:2410.05229.
- Roberto Navigli. 2009. Word sense disambiguation: A survey. *ACM Computing Surveys*, 41(2):1–69.
- Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*.
- Prisha Samadarshi, Mariam Mustafa, Anushka Kulkarni, Raven Rothkopf, Tuhin Chakrabarty, and Smaranda Muresan. 2024. Connecting the dots: Evaluating abstract reasoning capabilities of LLMs using the New York Times connections word game. In *Findings of the Association for Computational Linguistics: EMNLP 2024*.

- Shyam Sudhakaran, Miguel González-Duque, Matthias Freiberger, Claire Glanois, Elias Najarro, and Sebastian Risi. 2023. MarioGPT: Open-ended Text2Level generation through large language models. In *Advances in Neural Information Processing Systems*, volume 36.

- Graham Todd, Tim Merino, Sam Earle, and Julian Togelius. 2024. Missed connections: Lateral thinking puzzles for large language models. arXiv preprint arXiv:2404.11730.

- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35.

- Tianci Xue, Ziqi Wang, Zhenhailong Wang, Chi Han, Pengfei Yu, and Heng Ji. 2023. RCoT: Detecting and rectifying factual inconsistency in reasoning by reversing chain-of-thought. arXiv preprint arXiv:2305.11499.

- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L Griffiths, Yuan Cao, and Karthik Narasimhan. 2023a. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, volume 36.

- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023b. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

- Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc V Le, and Ed H Chi. 2023. Least-to-most prompting enables complex reasoning in large language models. In *International Conference on Learning Representations (ICLR)*.

## A Prompts (verbatim)

All prompts are stored as versioned text files under prompts/ in the repository and reproduced here verbatim.

### A.1 Stage 1 system prompt

You are an expert puzzle designer specializing in linguistic ambiguity. Always provide a complete Python list as requested.

### A.2 Stage 1 user prompts (tiers 1–4)

Each tier prompt shares the frame:

You are creating a Connections-style puzzle. Each group must contain exactly 4 related words or phrases.  
[tier-specific task, below]

569 Explain your reasoning step by step,  
 570 specifying the ambiguity type (polysemy,  
 571 homonymy, idiomatic, syntactic, figurative)  
 572 for each word or phrase. At the end, provide  
 573 the 4 final words/phrases as a Python list  
 574 in the format: ['word1', 'word2', 'word3',  
 575 'word4']. Ensure the list is complete and  
 576 correctly formatted.

## 577 Tier-specific tasks:

578 Tier 1: Group 1: Choose 4 words that be-  
 579 long to a clearly defined semantic category  
 580 (e.g., colors: red, blue, green, yellow).  
 581 Tier 2: Group 2: Choose 4 words with  
 582 slightly overlapping or flexible meanings  
 583 (e.g., tools: hammer, saw, drill, wrench).  
 584 Tier 3: Group 3: Choose 4 words  
 585 that exhibit polysemy (multiple related  
 586 senses, e.g., 'bank' as riverbank or fi-  
 587 nancial institution) or homonymy (same  
 588 spelling/pronunciation, distinct meanings,  
 589 e.g., 'bat' as animal or sports equipment).  
 590 Tier 4: Group 4: Choose 4 words or phrases  
 591 that exhibit idiomatic ambiguity (phrases  
 592 with non-literal meanings, e.g., 'break the  
 593 ice'), syntactic ambiguity (words that create  
 594 ambiguous sentence structures, e.g., 'saw' in  
 595 'he saw the man with a telescope'), or fig-  
 596 urative ambiguity (metaphorical meanings,  
 597 e.g., 'light' as brightness or metaphorical  
 598 hope).

## 599 A.3 Stage 3 (RCoT) prompts

### 600 System:

601 You are a puzzle auditor verifying semantic  
 602 ambiguity per a defined taxonomy.

### 603 User template ({word\_list} substituted):

604 You are verifying a word puzzle group.  
 605 The group contains the following words or  
 606 phrases:  
 607 {word\_list}  
 608 For each word or phrase, explain why it be-  
 609 longs in the group, focusing on its ambiguity  
 610 type (polysemy, homonymy, idiomatic, syn-  
 611 tactic, figurative). Describe how its multi-  
 612 ple meanings or interpretations relate to the  
 613 theme of ambiguity.

## 614 A.4 Solver prompts

615 The direct, CoT, and ToT solver  
 616 prompts used by the released har-  
 617 ness are in prompts/solver\_direct.txt,  
 618 prompts/solver\_cot.txt, and  
 619 prompts/solver\_tot.txt; the ToT prompt  
 620 explores three complete candidate partitions  
 621 (literal-semantic, wordplay-first, decoy-  
 622 averse), scores each against the validity  
 623 constraints of §4, and selects or backtracks.

## B Dataset details

Real puzzles. 1,114 official puzzles  
 (2023-06-12 through 2026-07-04; 4,456  
 groups) from the public NYT-Connections-  
 Answers archive, refreshed reproducibly by  
 data/update\_real\_data.py; the 830 puzzles  
 through 2025-09-19 carry per-group official  
 difficulty levels 0–3 and form the metric-  
 validation set (see §6.2). Single-prompt  
 synthetic set. 2,000 training / 400 test  
 puzzles generated with GPT-4o from a single  
 prompt without the ambiguity pipeline (6,016  
 unique parsed groups after deduplication and  
 cleaning). Pipeline corpus. 50 puzzles  $\times$   
 4 tiers with all stage-wise artifacts. Exact  
 file formats and regeneration commands are  
 documented in the repository README.

## C Human evaluation protocol (designed, not yet run)

Design. Within-subjects, three condi-  
 tions (real NYT / single-prompt GPT-4o /  
 pipeline), 6 puzzles per participant (2 per con-  
 dition, Latin-square ordered), target  $n \geq 30$   
 fluent English speakers recruited on a crowd-  
 sourcing platform with a standard Connec-  
 tions interface (four guesses allowed, one-away  
 feedback). Measures. Solve rate, groups  
 solved, time-to-solve, error count (behavioral);  
 post-puzzle 7-point ratings of difficulty, fair-  
 ness (“the intended groupings felt justified  
 once revealed”), and enjoyment; a forced-  
 choice “would you play another like this?”.  
 Analyses. Mixed-effects models with partic-  
 ipant and puzzle random effects; the pre-  
 registered primary hypotheses are (H1) be-  
 havioral difficulty ordering pipeline > single-  
 prompt > real, (H2)  $D(G)$  predicts behavioral  
 solve rate at the group level, (H3) pipeline puzzles  
 remain within one point of real puzzles  
 on fairness. H3 is the safeguard against the  
 failure mode where amplified ambiguity yields  
 unsolvable or arbitrary puzzles.